

TABLE I

CAPABILITY COMPARISON ACROSS METHOD FAMILIES (● = FULL, ◐ = PARTIAL, ○ = NONE). **SEM**: SEMANTIC (LLM/VLM) REASONING; **PROP**: ≥ 1 KHZ PROPRIOCEPTION IN THE REASONING LOOP; **ATTR**: VISION + PHYSICAL FAILURE ATTRIBUTION; **REC**: STRATEGY-LEVEL (B-CLASS) RECOVERY; **RFLX**: MILLISECOND REFLEX DURING DELIBERATION; **SAFE**: FORMAL SAFETY GUARANTEE; **MAP**: FAILURE-AWARE SEMANTIC MAP; **WB**: WHOLE-BODY LEGGED DYNAMICS. KiNO ALONE SPANS ALL EIGHT AXES.

Method	Sem	Prop	Attr	Rec	Rflx	Safe	Map	WB
<i>Adaptive locomotion (no semantics)</i>								
RMA [1]	○	○	○	○	○	○	○	●
DreamWaQ [2]	○	○	○	○	○	○	○	●
Load-Adaptive Loco. [3]	○	○	○	○	○	○	○	●
<i>Manip. failure attribution & recovery</i>								
REFLECT [4]	●	○	◐	◐	○	○	○	○
AHA [5]	●	○	◐	○	○	○	○	○
ARMOR [6]	●	○	◐	○	○	○	○	○
<i>Legged LLM / VLA navigation</i>								
QUAR-VLA [7]	●	○	○	○	○	○	○	●
NaVILA [8]	●	○	○	◐	○	○	○	●
QuadrupedGPT [9]	●	○	○	◐	○	○	○	●
Helpful DoggyBot [10]	●	○	○	◐	○	○	○	●
<i>Physically-grounded traversability</i>								
ProNav [11]	○	◐	◐	○	○	○	◐	●
VLM-GroNav [12]	●	◐	●	○	○	○	●	●
<i>Safe legged locomotion (formal)</i>								
Grandia et al. [13]	○	○	○	○	○	●	○	●
One Filter [14]	○	○	○	○	○	●	◐	●
KiNO (Ours)	●	●	●	●	●	●	●	●